

DCC Signal Decoder

User Guide

Version 1.0

Self-learning DCC accessory decoder for UK colour-light signals

This decoder is a self-learning DCC accessory decoder for UK-style colour-light signalling. It drives a 2, 3, or 4 aspect main signal head, plus an optional Junction Indicator, Position Light, and a second, fully independent signal head - all from a single board.

No knowledge of CVs, programming tracks, or decoder addressing conventions is needed. The decoder learns its own DCC address directly from your command station, and every other setting - signal type, brightness, routes, links between decoders - is configured through a straightforward browser-based settings page, reached over a USB connection. No app store, no account, no internet connection required.

This guide covers everything you can do from that settings page, plus how to get firmware onto the board in the first place using the companion Uploader application.

What you'll need

- The decoder board, wired into your signal hardware.
- A USB cable to connect the board to a Windows PC.
- The Uploader application (a single installer - see “Installing the Uploader” below).
- A DCC command station connected to your layout, for setting the decoder's address and operating it.

Quick Setup

The fastest path to a working signal, before diving into everything else this guide covers:

- Connect the decoder to power and to your DCC command station's track bus.
- Connect your signal to the decoder - R (Red), Y (Yellow), DY (Double Yellow), G (Green) and C (Common) are the terminals used for up to a 4 Aspect head. See “Board Connections” below for the full picture.
- Press and hold the button to enter Learn Mode.
- From your command station, send the accessory address you want this decoder to use. Done - the decoder is now addressed and running as a 2 Aspect signal by default.
- Want a different signal type instead? Enter Learn Mode again (hold the button), then send address 981 for 2 Aspect, 982 for 3 Aspect, or 983 for 4 Aspect - this takes effect immediately. Hold the button again to exit Learn Mode.

981/982/983 only work once the decoder already has an address - they're for changing the signal type on an already-set-up board, not for the very first address you teach it. Everything else - brightness, EX ports, Aspect Link, and more - is configured from the Board Settings browser page; see the rest of this guide.

1. Board Connections

Before the signal will work, it needs a power supply, a DCC track connection, and the signal head itself wired in - each on its own terminal block.

Power Supply

Terminal	Connects To
VIN	Positive from your power supply
GND	Negative / ground from your power supply

Use a dedicated DC supply for the board - do not connect this directly to DCC track voltage.

DCC Track

Terminal	Connects To
DCC TRACK	One wire from your DCC track bus
DCC TRACK	The other wire from your DCC track bus

Connect these directly across your DCC track bus, the same two wires you'd tap for any other DCC accessory decoder. Polarity doesn't matter - DCC is an AC-style signal and the decoder doesn't mind which way round the two wires go.

Signal Head

Terminal	Connects To
R	Red aspect cable
Y	Yellow aspect cable
DY	Double Yellow aspect cable
G	Green aspect cable
C	Common cable

- C (Common) is shared by every lamp in the signal head - connect it regardless of how many aspects your signal has.
- If your signal doesn't have a particular lamp - for example, a 2 Aspect signal has no Yellow or Double Yellow - just leave that terminal empty.

Double-check every cable is in the terminal it's labelled for before powering on - much easier to fix now than once the signal's mounted on the layout.

EX Ports (Optional)

General-purpose terminals for a Junction Indicator, Position Light, or second signal head, if you're using any of them - see "EX Ports" under Board Settings below. What each one does is decided in Board Settings, not by which cable you plug in; connect a cable to whichever terminal you've assigned that role to, and leave any terminal you're not using empty.

Aspect Link (Optional)

If you're chaining this decoder to another one for automatic block signalling (see “Aspect Link” below), a further terminal block connects this board to the next signal along the line:

Terminal	Connects To
LINK IN	LINK OUT on the decoder for the block ahead
LINK OUT	LINK IN on the decoder for the block behind

Only connect this if you're using Aspect Link. If you're not using that feature, leave both terminals empty - the decoder works perfectly well as a standalone signal without it.

No separate ground wire is needed for this link - every board already shares a common ground through the layout's DC power supply.

2. Installing the Uploader

The Uploader is a Windows desktop application that flashes firmware onto the decoder and gives you access to its settings page. It requires no other software - no Arduino IDE, no drivers downloaded separately, nothing typed at a command line.

- Download the latest installer.
- Run it and follow the prompts. When asked, leave the USB-serial driver checkbox ticked - this installs the driver needed for Windows to recognise the board over USB. It covers the great majority of boards; if Windows still doesn't see the board after installing, see “Troubleshooting”.
- Once installed, launch the Uploader from the Start Menu or your desktop.

The driver install requires administrator rights, which the installer will prompt for. This is normal and only needed once.

3. Uploading Firmware

The main Uploader window has three steps: choose a firmware version, choose the board's COM port, and click Upload.

Step 1 - Firmware Version

Pick the firmware version to install from the dropdown. Each version bundled with the Uploader can have its own User Guide and Change Log, opened with the buttons alongside the dropdown - use the Change Log to see what's changed before updating a board that's already running an older version.

Step 2 - COM Port

Connect the decoder to your PC with a USB cable. Its COM port should appear automatically in the dropdown within a few seconds; if it doesn't, check the cable and connection, then click Refresh. A red “No board detected” warning next to this step means Windows currently sees no serial device at all - this points at the cable, the port, or the driver, not at the decoder itself.

Step 3 - Upload

Click Upload Firmware. The Uploader will report progress in the Status log at the bottom of the window, and let you know when it's finished. Existing settings on the board are preserved across a firmware update wherever the Uploader is able to - you generally do not need to reconfigure the board from scratch after updating firmware, though it's always worth a quick check afterward that everything came back as expected.

If an upload fails partway through, unplug both the USB cable and the board's track/power supply, wait a few seconds, reconnect USB only, and try again. Most failed uploads are a timing issue on the board's side that a fresh power cycle resolves.

4. Setting the DCC Address (Learn Mode)

The decoder learns its own address directly from the command station - there's no separate programming step and no address to look up or type in anywhere.

- Press and hold the button on the decoder board. An LED will indicate the board has entered Learn Mode.
- From your command station, operate the accessory address you want this decoder to respond to (for example, throw or close the corresponding turnout/accessory address).
- The decoder captures that address, saves it, and leaves Learn Mode automatically.

From this point on, the decoder responds to that address for every normal DCC command. You can re-enter Learn Mode at any time to teach it a different address - the previous one is simply overwritten.

The current address (along with everything else) can also be checked at a glance on the Board Settings page, covered next.

What Address Does What

Once learned, your signal automatically claims however many consecutive addresses it needs, starting from the one you taught it. Using X as the address you sent:

Signal Type	Addresses Used	What Each One Does
2 Aspect	X	X Closed → Red X Thrown → Green
3 Aspect	X, X+1	X Closed → Red X Thrown → Yellow X+1 Closed → Yellow (no separate Double Yellow lamp) X+1 Thrown → Green
4 Aspect	X, X+1	X Closed → Red X Thrown → Yellow X+1 Closed → Double Yellow X+1 Thrown → Green

This table covers the main signal head only - it's all you need if you're operating straight from a command station with no computer involved. If you add a Junction Indicator, Position Light, or second signal head via Board Settings, each claims its own further address(es) immediately after these - Board Settings shows you the exact addresses once they're set up.

5. Board Settings

Board Settings is a configuration page built into the decoder itself, reached from the Uploader's main window by clicking the “Board Settings” button. It opens in your default web browser and works entirely over the USB connection - no network or internet access is needed, and nothing about the board's configuration is ever sent anywhere off your own PC.

Every setting on this page takes effect on the board immediately when you change it, and is saved automatically - there is no separate “Save” step for most controls, and no need to re-upload firmware to change a setting.

The page is organised into a small number of cards, described below.

Status

The top of the page shows the decoder's live state at a glance: its current DCC address, current aspect, occupancy state, and (if Aspect Link is in use) whether it's currently connected to its neighbours. This updates continuously while the page is open.

Signal Configuration

- Signal Type - choose 2, 3, or 4 aspect operation for the main signal head.
- Aspect Memory - when enabled, the signal remembers and restores its last aspect after a power loss. When disabled, it starts at Red instead (or at a Default Startup Aspect you choose, on firmware versions that offer it).
- End of Line - marks this decoder as the last signal in a sequence, with nothing wired ahead of it. Affects how Aspect Link derives this signal's aspect (see “Aspect Link” below).
- Allow Aspect Override - opt-in permission to manually or by DCC command move this signal to a less restrictive aspect while its block is occupied (for example, for a shunting move). Off by default as a safety measure - see “Aspect Link” for the full rule.

Not every option on this card appears on every firmware version - Board Settings only ever shows what your specific board actually supports.

Brightness

A global brightness dial scales every LED on the main signal head together, alongside individual sliders per LED colour for fine adjustment. If a second, independent signal head is mapped (see “EX Ports” below), it gets its own separate global dial and sliders, so the two heads can be balanced independently of each other.

EX Ports

EX ports are general-purpose outputs that can each be assigned one of several roles, letting one board do more than just drive a single signal head:

- Junction Indicator - shows which of several routes through a junction is currently set. Each available route combination gets its own DCC address; the indicator can be shown standalone or alongside the main signal.
- Position Light - a Closed/Thrown pair of indications, each independently addressable and switchable by its own DCC command.
- Second Signal - a fully independent 2, 3, or 4 aspect signal head, sharing the board but otherwise entirely separate from the main signal (its own aspect, its own brightness dial). Controlled directly by DCC command or manually from this page - it is not linked to occupancy detection or Aspect Link.

Changing an EX port's role only ever affects that port and, where a role claims more than one port (a multi-route Junction Indicator, or a Position Light's Closed/Thrown pair), the specific ports it needs. If a change would take over ports already claimed by something else, the page warns you first and shows exactly which ports will be cleared before it happens.

Aspect Link

Aspect Link lets decoders be chained together so that an upstream signal automatically reflects the state of the block ahead of it, for basic automatic block signalling - each board only needs to know about its immediate neighbour in each direction, not the whole line.

- Wiring: connect this decoder's OUT terminals to the next decoder's IN terminals in the direction trains travel, without crossing the pair - see "Board Connections" above. The Board Settings page shows a connectivity banner guiding you through this, and will flag in red if a link that was working has since gone quiet - a good first check if a chained signal stops updating.
- Occupancy: a signal only ever shows danger (or reflects an upstream danger) for its own block while that block is genuinely occupied. If you don't yet have a real occupancy sensor wired in, Board Settings includes a simulated occupancy toggle for testing the whole chain end-to-end before your track detection is in place.
- Manual/DCC override: while a block is occupied, a manual or DCC aspect change is only accepted if Allow Aspect Override is ticked for that decoder (see "Signal Configuration" above) - useful for shunting past a signal at danger with a train still standing in the block. While a block is clear, an override is capped at whatever aspect the block would show anyway based on the state of the block ahead - you can request something more cautious than that, but never something less cautious than the situation actually warrants.

A signal that loses contact with its neighbour mid-operation holds its last known aspect rather than defaulting to all-clear - it fails toward caution, not away from it.

Backing Up and Restoring Settings

Board Settings includes Export and Import buttons that save every setting to a single file on your PC, and can load it back later - useful for keeping a record of a working configuration, or copying one board's setup onto another.

Where possible, the Uploader also backs up a board's settings automatically before a firmware update and restores them once the update finishes, so a routine firmware update doesn't mean starting configuration over from scratch.

6. Reading the Light

A status LED on the board tells you what it's doing at a glance, without needing Board Settings open:

The Light Does This	It Means
OFF	Normal running - address learned, DCC present
Solid ON	Not yet learned - no address stored
Slow, steady flash (on/off)	Learn Mode - waiting for you to send an address
Two short blips, a pause, repeating	Address learned, but no DCC seen recently - check your track power and wiring
A handful of quick flashes	Saving a setting to memory - just wait a moment
Three flashes	Learn Mode finished successfully
A longer burst of rapid flashes	Starting Completely Over (factory reset) just happened

The status LED is a convenience for troubleshooting without a PC to hand - everything it tells you (current address, whether DCC is present, and more) is also always visible, in more detail, on the Board Settings Status card.

7. Starting Completely Over

If you ever want to wipe the decoder back to its factory defaults - every setting, every EX port role, everything - there are two ways to do it.

Via the button

- Hold the button down for about 10 seconds - noticeably longer than the 2 seconds used for Learn Mode.
- The light flashes rapidly to confirm.
- Every setting reverts to its factory default, including EX port roles - if you rely on a Junction Indicator, Position Light, or second signal, you'll need to set those up again afterwards in Board Settings.

Via Board Settings

Board Settings has a Factory Reset button that does exactly the same thing, without needing to reach the physical board.

Either way, the decoder is left fully usable immediately afterward - there's no separate step needed before it starts responding to DCC commands again.

8. Troubleshooting

No board detected / COM port doesn't appear

- Check the USB cable is a genuine data cable, not a charge-only cable, and try a different USB port or cable.
- Re-run the Uploader installer and make sure the driver checkbox was ticked.
- Open Windows Device Manager and look under “Ports (COM & LPT)” - if the board shows up with a warning icon, the driver didn't install correctly for that specific board.

Upload fails or gets stuck

- Unplug both USB and the board's track/power supply, wait a few seconds, reconnect USB only, and try again.
- Make sure Board Settings isn't open in a browser tab at the same time - close it before uploading, since the board's USB connection can only be used for one thing at a time.

Board Settings page won't load or shows stale data

- Close and reopen it from the Uploader - the page needs the Uploader's own bridge running in the background, so it won't work if the Uploader itself has been closed.
- Refresh the browser tab if data looks out of date - the page polls the board continuously, but a slow or dropped USB connection can occasionally leave it a step behind.

Aspect Link isn't working between two boards

- Check the connectivity banner on each board's Board Settings page - blue means never connected, red means it was working and has since gone quiet.
- Confirm OUT is wired to IN in the direction of travel on each cable, and that the pair isn't crossed.
- Confirm End of Line is only ticked on the actual last board in the chain - ticking it anywhere else stops that board from expecting a link at all.